PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013

Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

AI-Based Face Recognition Attendance Management System

A PROJECT REPORT

Submitted by

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Under the guidance of

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BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE

Certified that this report "**AI-Based Face Recognition Attendance Management System**" is a Bonafide work of "CSE-7302 Internship", who have successfully carried out the project work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE AND ENGINEERING during 2026.

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DECLARATION

We the students of final year BTech in COMPUTER SCIENCE ENGINEERING, at Presidency University, Bengaluru, named TEJASHWINI BV hereby declare that the project work titled "**AI-Based Face Recognition Attendance Management System"** has been independently carried out by us and submitted in partial fulfilment for the award of the degree of BTech in COMPUTER SCIENCE ENGINEERING, during the academic year of 2026. Further, the matter embodied in the project has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

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TEJASHWINI BV

Abstract

An AI-Based Face Recognition Attendance Management System is an advanced system that has been proposed to automate the attendance recording system in various organizations like schools, colleges, etc. Traditionally, various methods like attendance registers and cards have been used to record attendance. However, these methods are time-consuming, error-prone, and susceptible to various cheating practices like proxy attendance. Thus, in order to overcome these problems associated with traditional attendance systems, the proposed system has been developed using various advanced techniques of Artificial Intelligence, Computer Vision, and Machine Learning. This system uses face recognition technology to accurately identify an individual. For face detection, an image of the face is captured by the camera. This image is then compared with the pre-stored image in the database using face detection technology. Once the face is recognized, the attendance of the individual is automatically recorded in the system.

The system is also secure since the features of the faces are unique and cannot be easily copied. The platform can be used for the generation of attendance reports, monitoring of attendance patterns, and the provision of real-time updates. Through the integration of intelligent algorithms and automated management of the data, the AI-Based Face Recognition Attendance Management System provides a reliable, efficient, and effective solution for the management of attendance in the contemporary world. Not only is the system efficient in the management of attendance, but it is also contactless, and this is particularly useful in environments where hygiene is of utmost importance. Thus, the AI-Based Face Recognition Attendance Management System is a perfect example of how AI-based technologies can change the management of attendance from a traditional approach to a smart, secure, and automated system.

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Abbreviations

AI	Artificial Intelligence
CV	Computer Vision
API	Application Programming Interface
PKL	Pickle file format
DL	Deep Learning
SQL	Structured Query Language
SQLite	Structured Query Language Lite
CNN	Convolutional Neural Network
GUI	Graphical User Interface
CPU	Central Processing Unit
TTS	Text To Speech
TS	Timestamp
OS	Operating System

Chapter 1

Introduction

An AI-Based Face Recognition Attendance Management System is an automated attendance management system that uses face recognition technology for identifying people and marking their attendance without any physical contact or human intervention. The conventional attendance management systems, like maintaining attendance registers, ID cards, or biometric fingerprint-based systems, require physical contact between two individuals, time consumption, and sometimes even proxy attendance. However, face recognition-based attendance is faster, contact-free, and more secure.

The face recognition-based attendance management system works on the principle of acquiring images or video feed from a camera, identifying faces, and then matching them with a pre- stored database of people. By employing advanced computer vision techniques, face recognition systems can recognize individuals based on unique features like the distance between two eyes, nose shape, face structure, etc., and mark their attendance automatically upon identification.

1.1 Background

Attendance management is an important task in organizations, educational institutions, and working environments. Traditionally, attendance records are maintained manually using registers, sign-in sheets, and ID cards. However, such traditional methods are often found to be time-consuming and prone to human error and fraudulent practices such as proxy attendance.

Due to the recent advancements in Artificial Intelligence (AI) and Computer Vision, automated attendance management has gained popularity for efficient and accurate performance of tasks. Face Recognition is a subfield of AI and image processing that allows computers to recognize and identify faces based on facial characteristics. By recognizing the unique patterns of faces, AI can accurately recognize and identify faces even at different illumination conditions.

1.2 Statistics

The AI-Based Face Recognition Attendance Management System utilizes the advanced

computer algorithms based on Artificial Intelligence and Computer Vision techniques for automating the attendance recording process. Research suggests that traditional manual attendance systems require around 5-10 minutes per class, while face recognition attendance systems can record attendance in less than 30 seconds. This increases efficiency by around 80-90%. Statistical analysis of the attendance management system indicates that it improves accuracy, speed, and efficiency in attendance management compared to traditional attendance management practices.

1.3 Prior existing technologies

Prior to the advent of AI face recognition technology, organizations and institutions of learning employed various traditional methods of recording attendance. Among these was the traditional manual attendance register, which involved calling out students' names. Although simple, this method was cumbersome and often involved various human error issues, including proxy attendance.

Traditionally, before the advent of AI-based face recognition-based attendance management systems, various traditional methods of employee/student attendance tracking have been in use. The first such method was the use of employee/student attendance registers, whereby the employee/student had to record his/her attendance by simply signing the register.

1.4 Proposed approach

The proposed system utilizes the latest technologies in Artificial Intelligence and Computer Vision domains to provide efficient and automated attendance management using facial recognition technology. Images of the individuals are taken using a camera installed at the entrance of a classroom or a workplace. The images are then processed in real time using a face detection algorithm to detect and extract facial features from the images.

A database of the registered users is created during the enrollment phase. Images of each user are taken from different angles and stored in a database after converting the images into unique facial feature vectors. During the attendance marking process, the images are taken using a camera and matched against the facial feature database using a face recognition model.

1.5 Objectives

The project objectives are defined to be specific, measurable, and demonstrable:

1. **Automate Attendance Recording:** To automatically record attendance using face recognition technology, eliminating manual registers or biometric devices.
2. **Improve Accuracy:** To reduce errors such as proxy attendance, duplicate entries, and manual mistakes by using AI-based facial identification.
3. **Real-Time Identification:** To detect and recognize individuals in real time through a camera and mark attendance instantly.
4. **Enhance Security:** To ensure that only authorized and registered individuals are recognized by the system.
5. **Reduce Time Consumption:** To minimize the time required for taking attendance in classrooms, offices, or institutions.
6. **Maintain Digital Records:** To store attendance data in a centralized database for easy access, monitoring, and reporting.

1.6 SDGs

 Quality Education (SDG 4): Improves the management of student attendance.

 Industry, Innovation and Infrastructure (SDG 9): Makes use of AI technology for the smart management of attendance.

 Decent Work and Economic Growth (SDG 8): Helps organizations efficiently manage employees' attendance.

 Reduced Inequalities (SDG 10): Ensures that everyone makes use of the transparent attendance system.

 Sustainable Cities and Communities (SDG 11): Promotes smart management systems in institutions.

 Responsible Consumption and Production (SDG 12): Reduces the use of paper through the digital attendance system.

 Peace, Justice, and Strong Institutions (SDG 16): Ensures that the attendance system is secure and transparent.

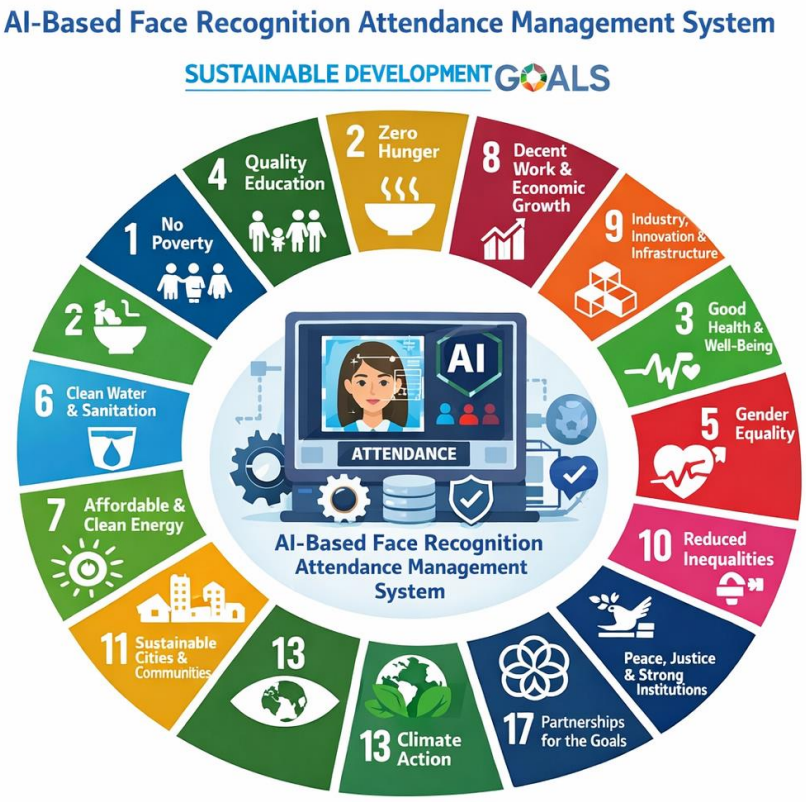

Fig 1.1 Sustainable development goals

1.7 Overview of project report

This report is organized into nine chapters. **Chapter 1** introduces the project, outlining the background, problem statement, objectives, and its alignment with the UN SDGs. **Chapter 2** reviews existing literature on employment portals and chatbot technologies. **Chapter 3** presents the methodology, detailing the system design and development approach. **Chapter 4** covers project management aspects such as the timeline, risk analysis, and budgeting. **Chapter 5** provides analysis and design, including requirements, block diagrams, flowcharts, and system modelling. **Chapter 6** describes the hardware, software, and simulation environment. **Chapter 7** discusses evaluation and results with test cases, insights, and validation. **Chapter 8** explores the social, legal, ethical, sustainability, and safety considerations of the project. Finally, **Chapter 9** concludes the report, summarizing contributions, highlighting limitations, and suggesting future work.

Chapter 2

Literature review

Shubh Agarwal., 2024 — Face Recognition-Based Attendance System- In this paper, an automatic attendance system based on face recognition technology was discussed. In this system, face detection was carried out using the Haar Cascade algorithm, and face recognition was performed using the LBPH classifier. Images of students are stored, and their attendance can be marked automatically. In this system, face detection was tested with 110 students, resulting in a face recognition accuracy of about 91%.

Arun Kumar Tiwari, Sneha Rai, Rehan Ahmad, 2025 — Face Recognition for Attendance Management System - The authors developed a chatbot to automate customer service operations using NLP and hybrid ML techniques. The system significantly reduced response time and lowered the load on human agents. Results indicated improved customer satisfaction due to quick and consistent replies. However, accuracy issues were observed for complex or multi-turn conversations.

Patel S., Kumar P., Garg S., Kumar R, 2024 — Face Recognition Based Smart Attendance System Using IoT- The system employs image processing and face recognition library to recognize the individuals and record the attendance in a database. The proposed system is implemented using a Raspberry Pi and a camera module to capture the images in real time. The proposed system also sends emails to the supervisor or parents if the person is absent. The proposed IoT-based system increases the accuracy, eliminates proxy attendance, and minimizes the time taken for the attendance mechanism.

Smitha, Pavithra S. Hegde, 2020 — Face Recognition Based Attendance Management System - The system recognizes the face by utilizing image processing techniques and comparing the image with the pre-stored data. It makes use of face detection through the Haar Cascade algorithm and face recognition through the LBPH algorithm to recognize the face. The attendance is tracked and stored electronically, reducing the need for manual intervention. It has been concluded that the system improves the efficiency, accuracy, and security of the attendance management system.

Shriyasti Shah, Keshav Sharma, Roopam Gupta, Manjot Kaur Bhatia, 2022 — Face Recognition Based Attendance Management System - The system uses a camera to capture images of the people. The system uses a comparison of the images taken to a database of facial images to identify people. The attendance of the people is then recorded in the database. The proposed system minimizes errors, proxy attendance, and increases the efficiency of the attendance.

Om Bhakat, Poulami Roy, Poushmita Paul, 2025 — Face Recognition Based Attendance Management System - The system uses a camera for image acquisition and computer vision for processing the images. It uses face detection and recognition techniques to match the faces captured by the camera with the stored images of registered members. After the faces are recognized, the system records the attendance digitally. The study concludes that the proposed system increases accuracy, saves time, and solves the proxy attendance problem of traditional attendance methods.

Agustiyar Agustiyar, R. Rizal Isnanto, Catur Edi Widodo, 2024 — Face Recognition for Attendance Systems - The study used the Scopus database to collect the required data from the years 2019 to 2024. The tools used for the analysis of the data included Open Refine and Biblioshiny. The results revealed an increasing trend of research on facial recognition attendance systems, especially from countries such as India, Indonesia, and Malaysia.

Shashidhar V et al., 2021 — Advanced Facial Recognition Attendance and Behavioral Feedback System - The system also utilizes face recognition technology to automatically recognize the students and record the attendance. Besides the recording of the attendance, the system also analyzes the student behavior during lectures using computer vision.

M. Varsha, S. Chitra Nair, 2021 — Attendance Management System Using Face Detection and Face Recognition - This system uses a camera to capture the images of the students. It uses image processing to identify the faces. It then uses the images to identify the faces and mark the attendance of the students accordingly.

Ashwin Rao, 2022 — A Real Time Attendance System Using Face Recognition - It employs deep learning face recognition models to correctly identify the person using the

database After the face is recognized, the attendance is automatically taken and recorded in a virtual database. This study proves the potential benefits of using real-time face recognition technology in enhancing the speed, accuracy, and reliability of the attendance management system.

Touzene Abderraouf et al., 2024 — n Embedded Intelligent System for Attendance Monitoring his includes the integration of face recognition technologies with hardware to recognize people in real time. The face recognition is achieved by taking a picture of the face using a camera and then using machine learning to recognize the face. The attendance information is recorded automatically in a database for easy accessibility. The research indicates that intelligent systems can be used to create efficient, affordable, and reliable solutions for tracking attendance.

Bao-Thien Nguyen-Tat, 2024 — Automating Attendance Management in Human Resources Using Computer Vision and Facial Recognition - The system is also designed to help in the improvement of workforce monitoring, reduction of manual attendance tracking, and prevention of time fraud. This study shows that computer vision-based attendance systems can help in improving efficiency and accuracy in HR management.

K. Susheel Kumar, Vijay Bhaskar Semwal, 2011 — Real Time Face Recognition Using AdaBoost Improved Fast PCA Algorithm - The proposed method helps in better feature selection as well as reducing complexity during the recognition process. Experimental results show that the system works efficiently in real-time environments with better recognition rates.

Adjabi et al., 2020 — Past, Present, and Future of Face Recognition - The study considers different approaches such as feature-based approaches, appearance-based approaches, and deep learning approaches. The importance of the improvement in face recognition accuracy using Convolutional Neural Networks (CNNs) is also discussed in the paper. Some challenges in face recognition, such as changes in illumination, pose, and facial expressions, are also discussed in the paper.

Li Li, et al., 2020 — A Review of Face Recognition Technology - The study explores face

recognition techniques, which include traditional feature-based approaches and deep learning methods. It also points out the efficiency of Convolutional Neural Networks (CNNs) in enhancing the accuracy and performance of face recognition systems. It also identifies some challenges that face recognition systems, which include illumination, pose, and occlusion. The review concludes that face recognition systems have been significantly enhanced by the use of deep learning methods.

A. Kumar and M. Kumar, 2019 — Face Detection Techniques - The paper also discusses the traditional methods such as Haar Cascade, Principal Component Analysis (PCA), and AdaBoost face detection algorithms for the purpose of face detection in images. The paper also talks about the advantages and disadvantages of these methods in varying situations such as lighting and pose changes. The paper also explains the importance of face detection as a first step towards face recognition.

Based on the existing research, there is a need to develop an AI-based face recognition attendance management system, as it will increase the efficiency of the attendance management system for educational institutions and workplaces. It allows for a secure and touchless way of recording attendance without any issues of proxy attendance or time fraud. The gaps that have been identified in the existing research works are the integration of face recognition in real time, database management, and the use of AI for improving the accuracy of the face recognition-based attendance management system for better performance in varying lighting conditions and for a larger number of users.

Chapter 3

Methodology

3.1 Methodology Adopted

The Software Development Life Cycle (SDLC) methodology employed in the development of the proposed system, "AI-Based Face Recognition Attendance Management System," is a systematic approach to ensure efficient system development and deployment. Initially, in the "Requirement Analysis" phase, the system requirements were identified, including automation of attendance marking, real-time face recognition, database storage, and security against unknown users. In the "System Design" phase, the system architecture was designed, including modules for face detection, encoding, real-time recognition, database handling using SQLite, and image capture storage. Furthermore, in this phase, the system design also includes the inclusion of various libraries, including OpenCV for image processing and InsightFace for deep learning-based face recognition. In the "Implementation" phase, programming using Python is employed to create modules for extracting facial features, comparing these features with existing features, and automatically logging attendance with timestamps.

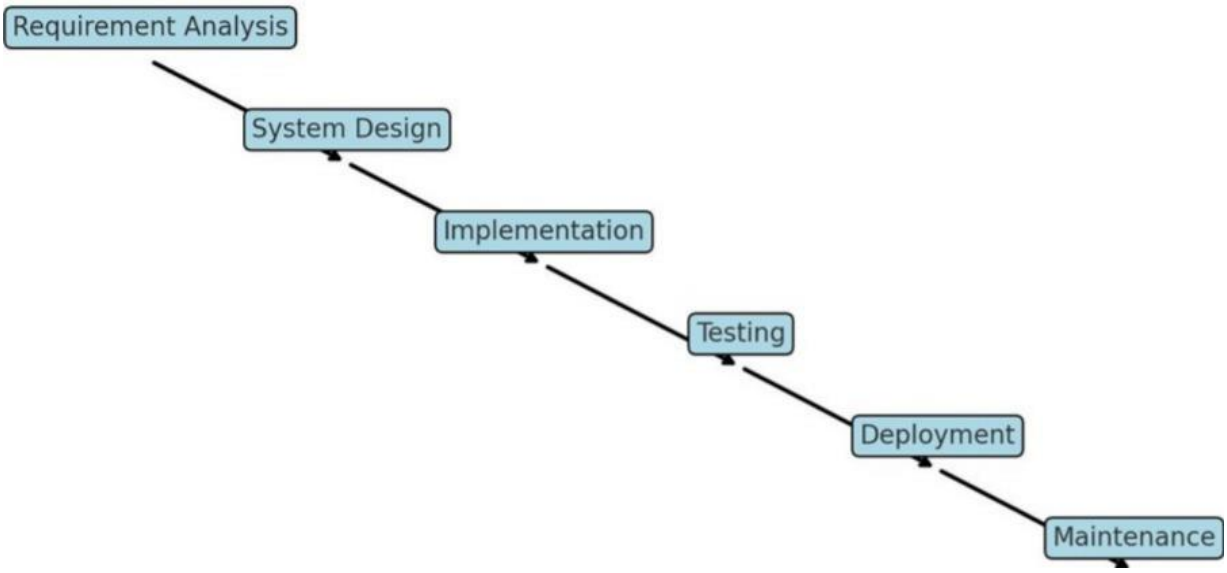

Fig 3.1 SDLC Methodology

3.2 Mapping Project Stages to SDLC

1. Requirement Analysis

The purpose of this phase was to understand the system expectations and define clear objectives.

Problem Identification

Traditional attendance systems used in schools and organizations have several limitations:

- Proxy attendance is also a possibility.
- Manual system is prone to human error.
- It is not easy to maintain and analyze attendance data.
- It is not a real-time monitoring system.

Objectives Finalized

- Automated detection and recognition of faces using Artificial Intelligence.
- Eliminate manual work and human error.
- Prevention of proxy attendance.
- Storage of captured images for verification purposes.
- Voice notifications for verified users.

Research and Literature Review

Face recognition involves the use of deep learning and computer vision techniques to identify people based on their facial characteristics.

Modern face recognition techniques involve the following:

- Convolutional Neural Networks (CNNs).
- Deep Learning-Based Embeddings.
- Feature Extraction and Comparison.

2. System Design

This phase translated the requirements into architectural and functional blueprints.

High-Level System Architecture

The system consists of the following components:

1. **Input Layer** – Webcam captures live video frames.
2. **Processing Layer** – Embeddings compared with stored embeddings.
3. **Data Management Layer** – Captured image path.
4. **Output Layer** – Displays recognized names on screen.

Why MongoDB

- **Flexible Schema -** MongoDB has flexible schemas that allow dynamic data structures without any fixed table.
- **Scalability -** Easy to manage huge data related to attendance from multiple devices.
- **High Performance -** Helps to perform read and write operations at high speed.
- **Image Metadata Storage -** Helps to store image paths as document format.

Functional Workflow

1. System starts and loads stored face embeddings.
2. Webcam takes video frames.
3. InsightFace detects faces and extracts face embeddings.
4. Embeddings are compared with stored embeddings using cosine similarity.
5. If match found → Display the name of the person and a welcome message.
6. If no match found → Display as Unknown Person and alert.
7. Attendance information (name, date, time, image) stored.
8. System continuously monitors until user exits.

Interface and Experience Design

- **Real Time Video Interface** - Displays webcam feed with face bounding boxes and user names.
- **Visual Feedback** - Displays green text for recognized persons and red text for unknown persons.
- **Voice Interaction** - Greetings to users with text-to-speech welcome message.
- **Security Alerts** - Beep alarm notification when unknown persons appear.

3. Implementation

Implementation involved converting design specifications into working modules.

Frontend Development

- Use OpenCV's GUI window to display the live camera feed and the faces that were detected.
- Show the bounding boxes and the name of the face on the screen.
- Display messages like "Welcome User" or "Unknown Person".

Backend Development

- Develop the face embedding generation module to encode the faces present in the dataset.
- Implement the face recognition logic to match the faces present in the dataset using the cosine similarity method.
- Design a SQLite database schema to store the attendance records with the name of the person, date, time, and the path of the images.
- Develop a logging mechanism to store the images of the persons in their respective folders.
- Develop the logic to prevent duplicate attendance within a certain time interval.
- Develop the logic to include voice notifications and alarm notifications for recognized persons as well as unknown persons.

Chatbot Development

- Utilize InsightFace (Buffalo_L model) to perform accurate face detection and embedding extraction.
- Train the system by encoding multiple images per individual to derive accurate embeddings.
- Employ cosine distance thresholding to establish accurate matches in identity.
- Improve system accuracy by increasing the number of images in the training set.

Version Control

- Use the Git tool to manage the source code and keep track of any changes made to the project.
- Create different branches to carry out the development process, the testing process, and the production process.
- Create a GitHub repository to store the code of the project to make the process easier.

4. Testing

Testing ensured correctness, reliability, and quality of the system.

Unit Testing

- Individual modules like face detection, embedding generation, database insertion, and image capture are being tested.
- The embedding generation module is being verified to check the correct encoding of the face from the dataset images.
- The database module is being tested to check the correct insertion of the attendance records into the SQLite database.
- Error handling is being checked for conditions like the absence of dataset folders, unreadable images, or no faces detected.

Integration Testing

- Verify that the embeddings match the stored embeddings during the face recognition process.
- Verify that the image capturing system saves images to the correct folder structure.
- Integrate the voice alert system with the ‘pyttsx3’ library and the alarm sound for unknown persons.
- Verify that the face bounding boxes and face recognition labels are correctly displayed on the video feed.

UI/UX Testing

- Verify whether the faces corresponding to the bounding boxes and the name are correctly made visible on the screen of the live camera.
- Verify whether the screen is correctly showing the messages like 'Welcome [Name]' and 'Unknown Person'.
- Verify whether the interface is smooth when faces are being detected.

Performance Testing (Basic)

- Test the face detection and recognition speed of the system using real-time video streaming.
- Test the response time of the system when dealing with multiple faces at the same time.

5. Deployment (Prototype)

- The developed face recognition attendance system can run on a local computer with a webcam
- The trained face embeddings and the attendance database can be integrated with the face recognition attendance system.
- Testing of the face recognition attendance system can be conducted with a small number of users, typically within a classroom or office setting.
- Feedback can be obtained from users to make improvements before scaling up the face recognition attendance system.

6. Maintenance and Future Enhancements

Even after prototype development, the system is structured for scalability.

Planned Future Enhancements

- Add the facility of cloud database support to enable the central monitoring of attendance records through various devices or locations.
- Create a web or mobile interface for the administrator to view attendance reports, analytics, etc.
- Add the facility of multi-camera support to monitor different entry points simultaneously.
- Enhance the accuracy of the attendance recognition system by using more data and better face models.

Maintenance Tasks

- Regularly update the face dataset and retrain the embeddings when new users are added.
- Database performance should be optimized and monitored to clean up any unnecessary data.
- Software dependencies like OpenCV, InsightFace, and Python should be updated.
- Backup and recovery of the database and attendance images should be maintained.

3.3 Verification and Validation

Verification and validation guarantee that the AI-based face recognition attendance system is correctly implemented and accurately performs face detection, face recognition, and attendance as required.

Verification Activities

- Verification of the source code was performed to ensure that face detection, face embedding, and database storage were correctly implemented.
- Individual face detection, face embedding, and database storage were performed to verify that the individual modules were working correctly.
- File paths and database connection were performed to verify that the program was running without any runtime errors.

Validation Activities

- The system was also tested with various images of faces to confirm the correct recognition of registered users.
- Real-time camera testing was carried out to test the correct detection of faces and attendance marking.
- The system was also tested with unknown faces to confirm the correct identification of faces.
- Attendance records were verified from the SQLite database to confirm the correct date, time, and image capture.

Verification and validation were carried out to ascertain the reliability and correctness of the AI-based face recognition attendance system. Verification was mainly carried out to ascertain whether the AI-based face recognition attendance system's constituent parts, such as data set processing, face embedding generation, and database integration, were implemented as per the specifications.

Chapter 4

Project Management

4.1 Project timeline

The project was executed in two main phases: **Planning** and **Implementation**. Each phase was divided into specific tasks with weekly deadlines. The Gantt charts below illustrate the sequence and overlap of tasks.

Project Planning

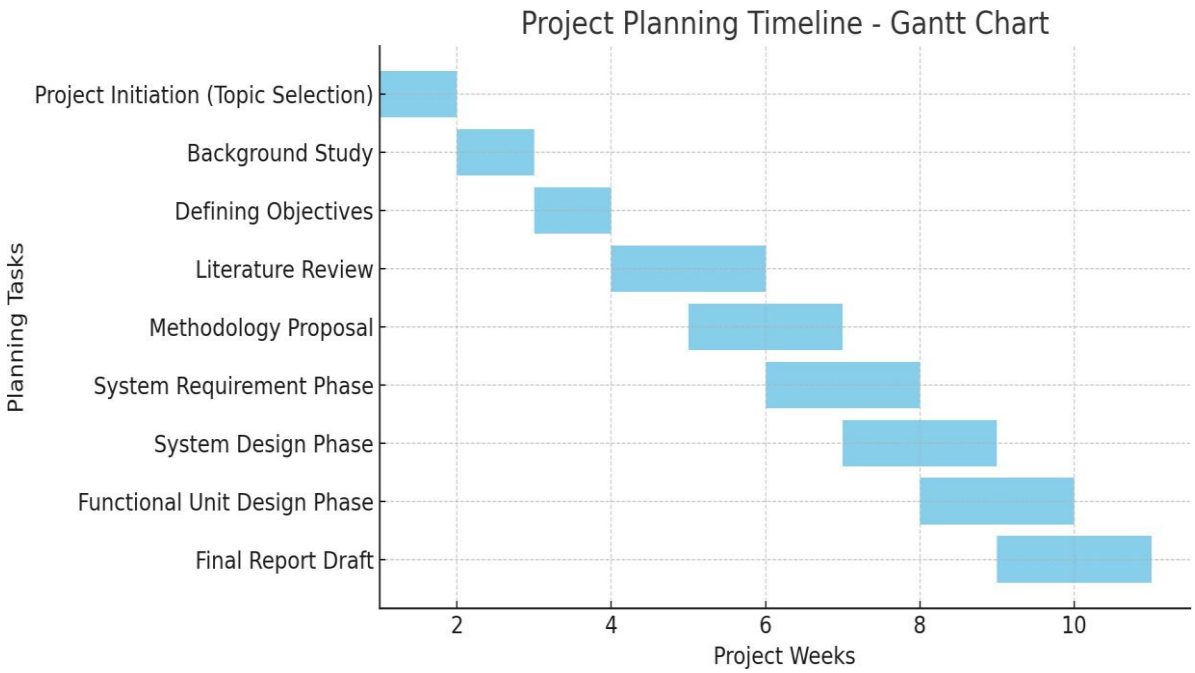

Fig 4.1 Project planning timeline

A. Planning Phase (Weeks 1–10)

Tasks & Explanations

1. **Project Initiation (Topic Selection)** – Week 1
    - Finalize the topic for the project, and then seek approval from the supervisor.
2. **Background Study** – Week 2
    - Conduct research regarding the existing attendance systems and the face recognition technology.
3. **Defining Objectives** – Week 3
    - Establish the goals for the face detection, recognition, and logging.
4. **Literature Review** – Weeks 4–5
    - Conduct research on at least 10 papers regarding the face recognition technology and attendance automation.
5. **Methodology Proposal** – Week 5–6
    - Decide upon the development methodology for the project, such as the waterfall or agile method, and design the diagrams accordingly.
6. **System Requirement Analysis** – Week 6–7
    - Establish the functional and non-functional requirements for the system.
7. **System Design** – Week 7–8
    - Design the architecture, UML, and DFD diagrams for the system.
8. **Functional Unit Design Phase** – Week 8–9
    - Code the system, integrate the modules, and then test the system accordingly.
9. **Final Report Draft (Initial)** – Week 9–10
    - Prepare the report, results, and presentation/demo.

Project Implementation

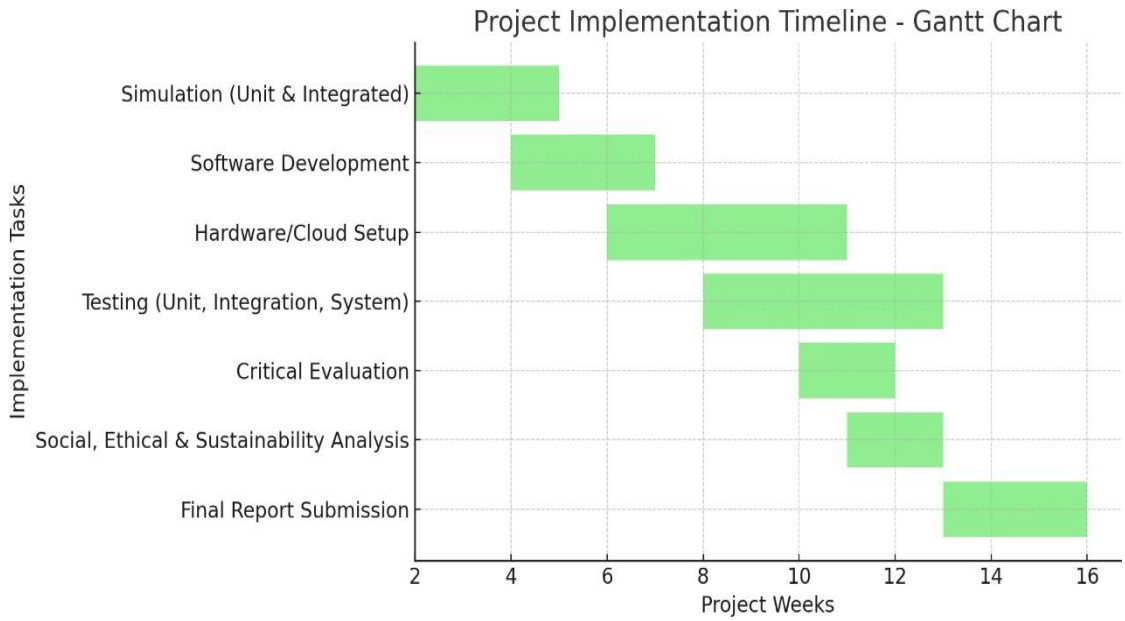

Table 4.2 Project implementation timeline

B Implementation Phase(Weeks2-15)

Tasks & Explanations

1. **Simulation (Unit & Integrated)** – Weeks 2–4
   - Simulated face detection & recognition modules with test images.
2. **Software Development** – Weeks 4–6
   - Developed GUI & backend for live attendance logging.
3. **Hardware/Cloud Setup** – Weeks 6–10
   - Configured the camera hardware, database, & cloud storage for captures.
4. **Testing (Unit, Integration, System)** – Weeks 8–12
   - Conducted unit, integration, & system testing for the face recognition & attendance.

5. **Critical Evaluation** – Week 10–11
    - Conducted a feasibility analysis for accuracy, speed, & reliability.
6. **Social, Ethical & Sustainability Analysis** – Weeks 11–12
    - Conducted a feasibility analysis for ethical, privacy, & sustainability issues.
7. **Final Report Submission & Presentation** – Weeks 13–15
    - Develop a final report, Diagrams & Presentations for the System Implementation.

4.2 Risk analysis

- **Privacy Breach:** Storing images and facial embeddings may lead to the breach of personal information if the database or files are compromised.
- **Unauthorized Access:** If the system or the database is not secured, the perpetrator may manipulate the attendance records.
- **False Positives/Negatives:** Facial recognition may lead to incorrect attendance records or failure to recognize the person.
- **Bias in Facial Recognition:** Facial recognition may not work well for particular demographics, leading to biased attendance records.
- **Data Loss**: Images captured may be lost, as well as the information stored in the database, if the system is not backed up.
- **Hardware Dependency**: Malfunction of the webcam may stop the attendance recording process.
- **Malware / Tampering**: A hacker may replace images, embeddings, or add fake faces to the database for malicious purposes.
- **Legal Compliance**: Facial recognition may violate local laws regarding the use of such information without consent.

- **Overhead / Scalability:** Facial recognition may lead to slower responses for large datasets, affecting real-time attendance recording.
- **Audio/Alarm Disturbance**: Continuous voice greetings or beeps may disturb the environment or violate workplace policies.

4.3 Project budget

The project demands investment in software as well as hardware components. Some of the major costs include investing in the purchase of the camera for high-resolution images, investing in a computer with a suitable GPU for efficient use of the InsightFace software, and investing in the purchase of the required storage devices for the datasets as well as the images for attendance. Other costs may include investing in software, investing in the purchase of a database management system, investing in the purchase of voice as well as alarm systems, investing in the time required for coding the application, and miscellaneous costs.

Budget Components

1. **Hardware:** High-resolution webcam(s), GPU-enabled PC/laptop, storage drives.
2. **Software & Libraries:** Python libraries (OpenCV, InsightFace, pyttsx3), optional commercial licenses.
3. **Database & Storage:** SQLite database, cloud or local storage for images and embeddings.
4. **Labor Costs:** Developer/programmer hours for implementation, testing, and debugging.
5. **Miscellaneous**: Electricity, backup solutions, alarms/speaker system, etc.

Chapter 5

Implementation

5.1 System Implementation

The AI-Based Face Recognition Attendance Management System works by using a camera-enabled device to obtain real-time face images of the user. The face recognition attendance management system works by detecting faces using computer vision and then extracting face features for recognition. The face features are then compared to a pre-trained database of face images of registered users. If the face features are found to be a match, then the attendance of the user is automatically registered in the database along with the date and time.

Dataset Folder:

- Contains all registered users' face images organized (often one folder per person).
- Used as the training source for the system to learn facial features.
- Images should have clear faces with different angles and lighting.
- Each image is labeled with the person's name or ID.
- The face recognition model reads these images to generate face encodings.

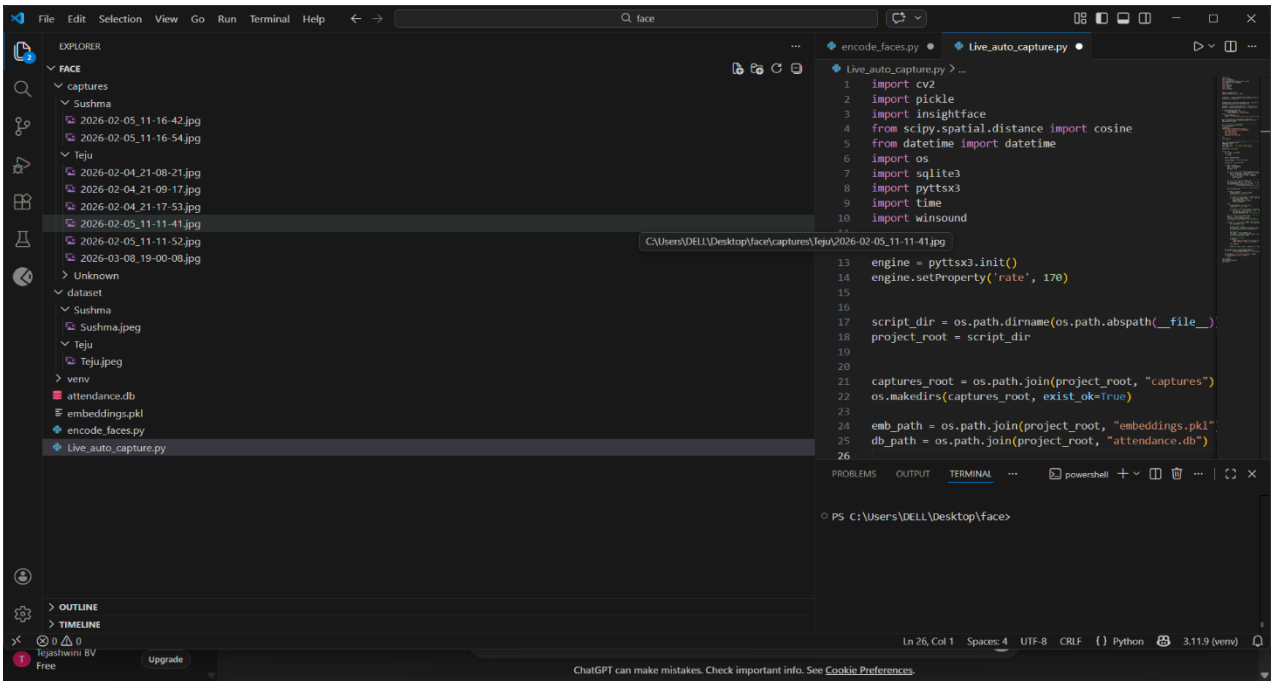

Fig 5.1 Dataset Folder

Face Encoding Module

- Detects faces from images or camera frames using a face detection algorithm.
- Extracts unique facial features such as eyes, nose, and face structure.
- Converts the detected face into a numerical vector representation.
- Uses libraries like OpenCV or face recognition to generate encodings.
- These encodings are used to compare and identify faces during attendance.

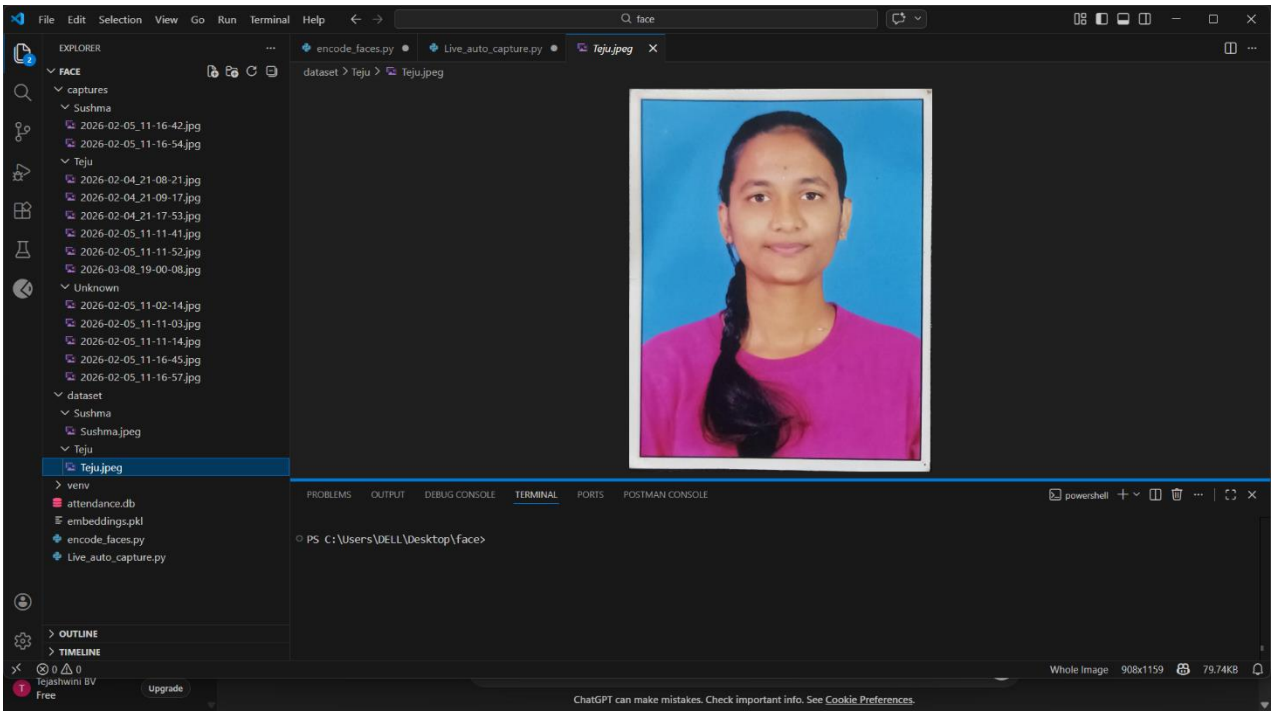

Fig 5.2 Face Encoding Module

Stored Face Vectors

- Numerical representations (usually **128-dimension vectors**) of each person's face.
- Generated from the dataset images during the encoding process.
- Stored in files or databases along with the person's name/ID.
- During recognition, the system compares live face vectors with stored vectors.
- If similarity distance is below a threshold, the person is **recognized and attendance is marked**.

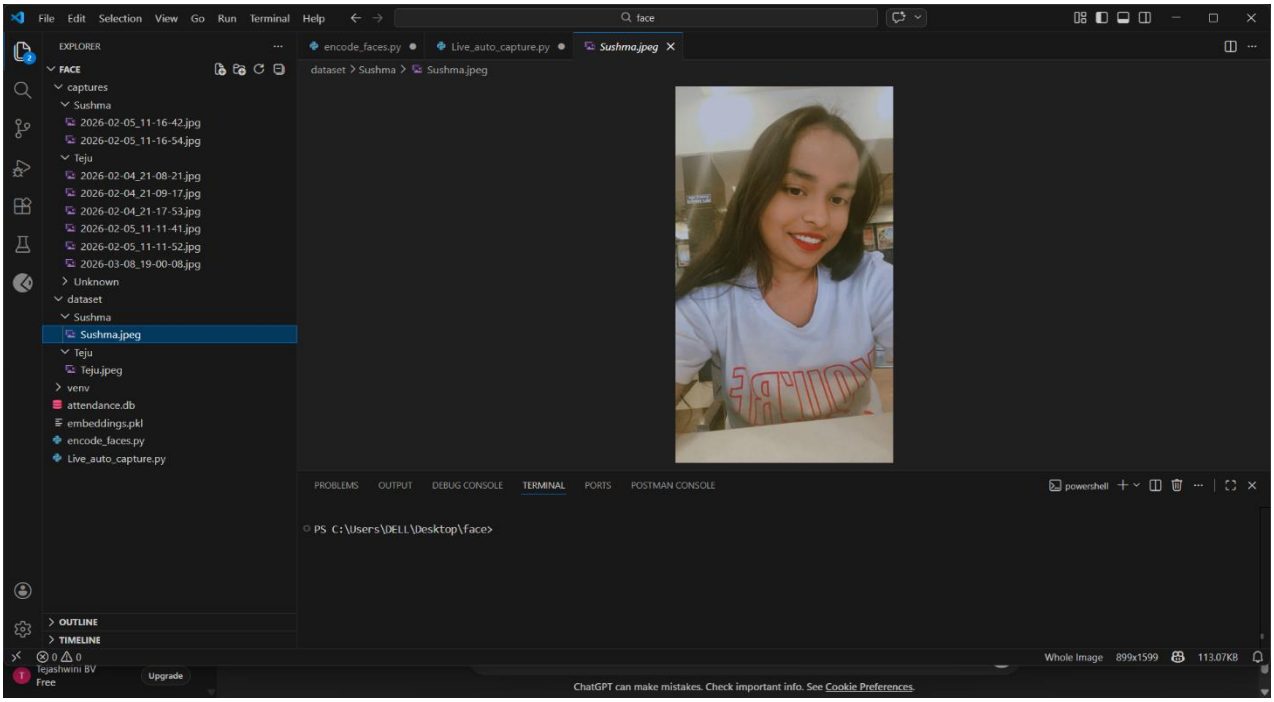

Fig 5.3 Stored Face Vectors